Measurement Without a Trusted Instrument

Hasok Chang's epistemic iteration and Eran Tal's model-based measurement, applied to AI introspection

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Measuring AI Introspection

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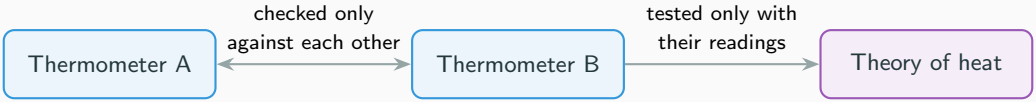
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Part 1

The problem: no trusted thermometer

A thermometer with nothing to check it against

How do we know a thermometer reads the temperature correctly? The obvious test is to compare it with a thermometer we already trust. Early thermometry had no such thermometer.

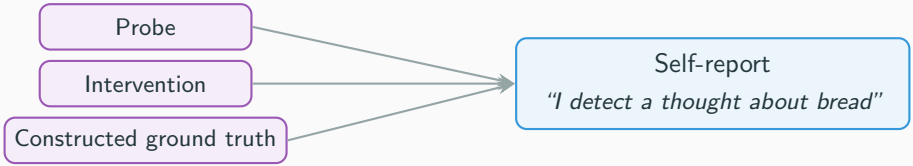


Every way of checking the instrument seems to need an instrument that does not yet exist. Hasok Chang's *Inventing Temperature* (2004) is the history of how thermometry was built anyway. Eran Tal's model-based account explains what calibration does when there is no independent access to the quantity.

The same situation in a language model

The AI version is the **concept-injection** experiment:

1. Take a vector for a word such as *bread*.
2. Add it to the model's activations while it answers.
3. Ask the model whether it detects an injected thought, and what it is about.



No instrument for reading a model's internal states is already trusted. Each instrument the report can be checked against is itself still being developed.

Three answers

Three bodies of work answer the question of how to trust an instrument without an independent ground truth.

Account	What does the justifying
Model-based (Tal)	The model of the measurement process. An outcome is inferred from an indication through that model, and trust comes from coherence across a network of instruments.
Iteration (Chang)	The history of corrections. Start from a standard accepted without proof and correct it through the inquiries it makes possible.
Robustness	The independence of the procedures. Agreement among independent procedures is evidence that a result is real.

All three agree that no measurement is justified from a foundation outside all theory. This deck covers the first two.

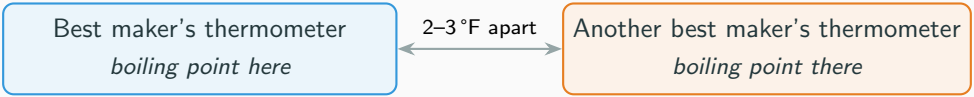
Part 2

Chang: building temperature

Fixed points

To mark a scale on a thermometer you need **fixed points**: phenomena that always happen at the same temperature.

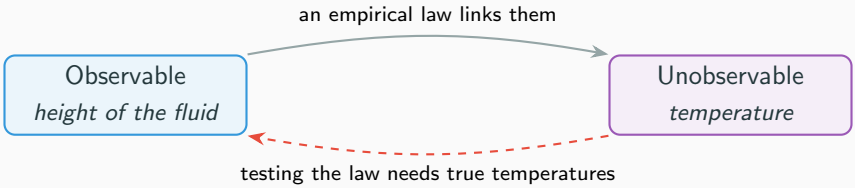
Early thermometry had none that were agreed on, and no way to check a proposed fixed point without a thermometer already trusted.



Even boiling water was not fixed. In 1776 a Royal Society committee found that thermometers from the best makers disagreed on the boiling point by 2–3 °F.

The problem of nomic measurement

Thermometers of mercury, alcohol and water, calibrated at the same fixed points, disagree. Which fluid expands uniformly?



Chang generalizes: when a measurement rests on a law linking an observable to an unobservable quantity, testing the law needs the very measurement it is supposed to justify.

Tal's encyclopedia entry places it beside van Fraassen's **problem of coordination**: which procedure measures a quantity and what the quantity is cannot be settled separately.

Comparability

In the 1840s Regnault tested thermometers without assuming any theory of heat. His criterion was **comparability**: a thermometer that reads the true temperature must at least give the same reading in the same circumstance, and all thermometers of one type must agree with each other.

Fails

Mercury thermometers made with different glasses disagree.

Passes

Air thermometers of different densities agree.

Regnault never treated comparability as truth: air thermometers passing did not show that they read the true temperature.

The principle of single value

Chang names the assumption behind comparability: *"a real physical property can have no more than one definite value in a given situation."*

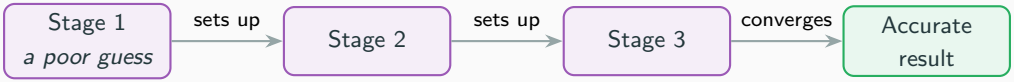
The principle is neither logical nor empirical, and no observation could test it. Regnault's test was **minimalist**: he compared thermometers of one type against each other and added no hypotheses about heat, so a failure could be blamed only on the thermometers.



The limit: several types can each pass the comparability test and still disagree with each other. Minimalism strengthens a verdict when there is one, but cannot guarantee a verdict.

Epistemic iteration

Chang's answer to the circle: start from an existing system of knowledge, accepted without proof that it is correct, and use it to run inquiries that refine and correct it.



Chang's example is a reconstruction of how Fahrenheit could have set the temperature of his mixing vessel: start from a poor guess and let each trial set up the next, converging on an accurate result.

The circularity cannot be escaped, so justification in empirical science has to be coherentist. What justifies the result is the self-correcting progress, not a secure foundation.

Two kinds of progress

Enrichment

The starting system is refined, not rejected.

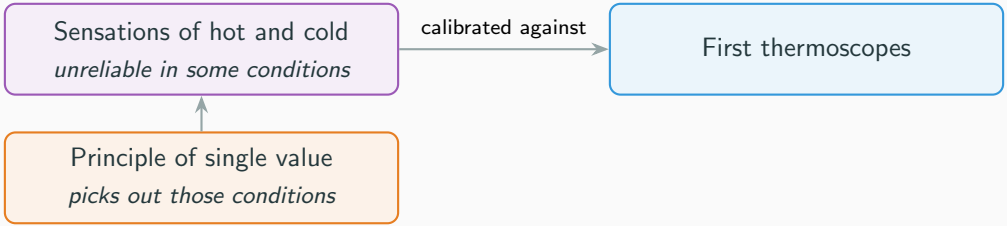
Self-correction

The starting system is changed by inquiry based on itself. Assuming one type of thermometer was correct produced observations showing that thermometers of that type disagreed.

- Nothing guarantees that iteration succeeds. If every attempt to improve a system contradicts itself, the system has failed.
- Competing traditions can progress in parallel, each on its own basis.

Calibration against an inaccurate standard

An objection: a procedure cannot be calibrated without an accurate one. Matthias Michel (2021) answers it with Chang's history.



The first thermoscopes were calibrated against human sensations of hot and cold, which were known to be unreliable in some conditions. The principle of single value picked out those conditions. Inaccurate procedures are not useless.

Metrological extension

Temperature measurement was carried into extreme cold and heat, where mercury froze and glass melted. Chang calls carrying a measurement method into a new domain **metrological extension**.

The method is valid if it coheres with the rest of the concept's meaning. Two conditions:

Conformity

If the concept already has a meaning in the new domain, the new standard should agree with it.

Overlap

If the old and new standards share a domain, they should give consistent results there.

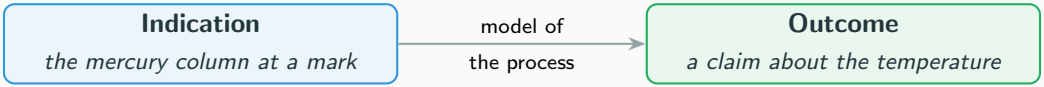
Michel (2024) adds an **existence condition** (the construct must be present in the new domain) and requires the shared domain. When the conditions fail, a validity drift can follow.

Part 3

Tal: measurement as inference

Indication and outcome

Tal's measurement has two levels: a concrete process in which object, instrument and environment interact, and an abstract model of that process.



- An **indication** is a property of the instrument in its final state: a pointer position, a number, a symbol or any other output.
- An **outcome** is a knowledge claim about the object, inferred from indications together with background knowledge.
- Without the model there is no outcome. The same indications can support different outcomes under different models.

Three kinds of calibration

For Tal (2017), calibration is modeling: building models of the measurement process, deriving predictions and testing them.

Black box
one input-output unit

Outcomes in many circumstances, low accuracy

White box

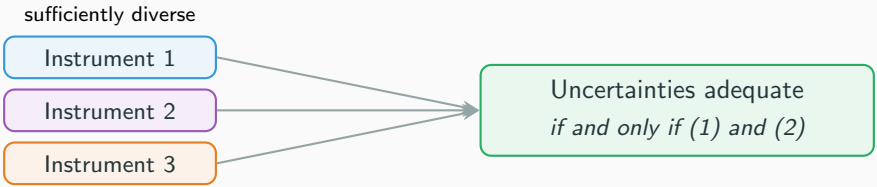
measurand | parts | object | environment | operators | data processing

Accurate, but carries over only within the narrow conditions it models

Coherent calibration models the process so that it fits background knowledge and applies the quantity consistently across objects, procedures and conditions. Black-box and white-box calibration are special cases of it.

The generalized robustness condition

Calibration cannot test outcomes independently of every background assumption. It can test them across a network of instruments.

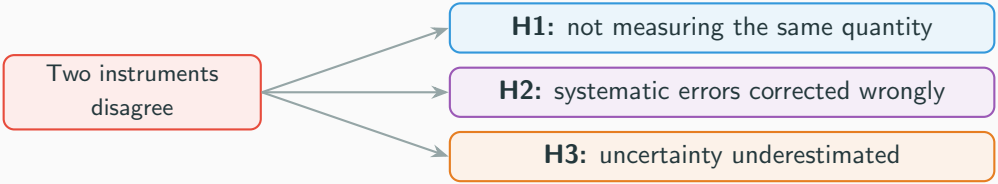


1. Their outcomes agree within those uncertainties.
2. Each uncertainty comes from an adequate model of its process.

Calibration cannot verify outcomes independently of any model, but it can verify them independently of any *particular* model, instrument and environment.

Quantity individuation

When do two procedures measure the same quantity? Tal (2019): when their outcomes disagree, three explanations compete.

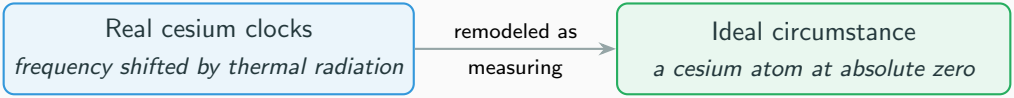


These cannot be tested one at a time. Establishing what instruments measure, how accurate they are and whether they agree are entangled tasks, and no context-free rule settles the choice.

Tal's criterion: two procedures measure the same quantity if both can be modeled coherently in terms of the same type of quantity in the background theory.

The cesium clock

One route to that criterion is **unity through idealization**: different concrete circumstances measure the same quantity if they are modeled as approximations of the same ideal circumstance.



When thermal radiation was found to shift the clocks' frequency, they were remodeled as measuring the frequency of a cesium atom at absolute zero.

On this view, systematic error is what lets different instruments be modeled as measuring one quantity despite stable disagreements among their readings.

Part 4

AI introspection

Tal's terms in an introspection experiment

Model-based term	In a concept-injection experiment
Indication	The model's reply: yes or no, a named concept, a confidence number, a free-text description
Outcome	The claim that the model has access to, or represents, a given internal state
Model of the process	Assumptions about prompting, sampling, the judge, the intervention, and the route from internal state to reply
Black-box calibration	Fitting reply rates to a constructed ground truth, such as injected versus not injected, without modeling how the reply is produced
White-box calibration	Modeling the report pathway itself, for instance by mechanistic analysis of the circuit that produces the reply
H1 / H2 / H3	When a self-report and a probe disagree: different quantities, uncorrected bias, or understated uncertainty

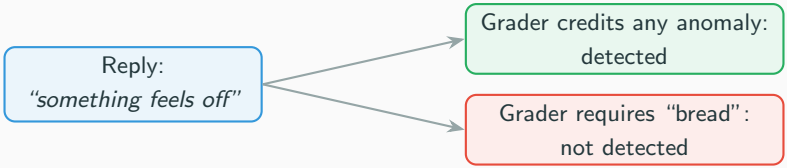
Scoring and scope

Scoring is part of the model

The same indications support different outcomes under different models. A grader that credits any report of an anomaly and a grader that requires the correct concept turn the same replies into different outcomes.

A calibration holds where it was done

A calibration against injected vectors at one layer and strength carries over, on Tal's account, only within the conditions under which it was made.



Michel calls the grader's choice the criterion content.

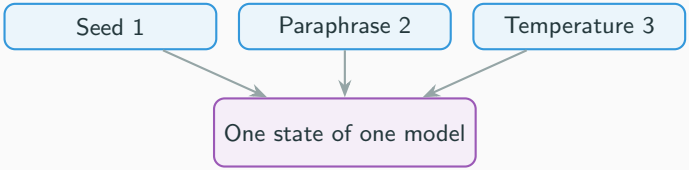
Fixed points and comparability in a model

Fixed points

A paper that uses injected vectors as its reference assumes the injected state is the same across layers, strengths and prompts. Chang's boiling point is the case in which an assumed fixed point turned out to depend on unspecified conditions.

Comparability as a minimal check

Runs of one report procedure under different seeds, paraphrases or temperatures are thermometers of one type. The principle of single value requires that they agree about one state of one model.

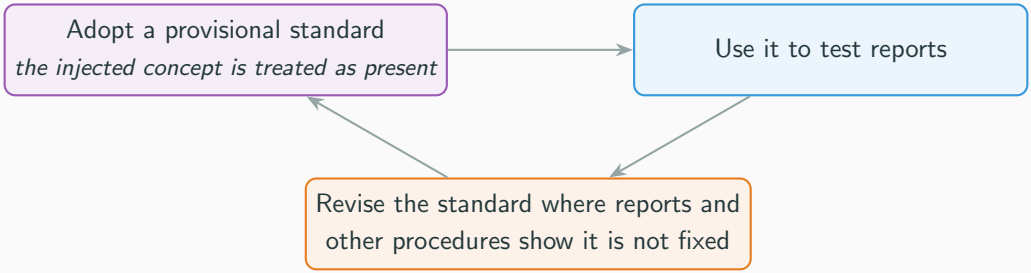


Fail: ruled out without any theory of the state.

Pass: not thereby validated, because several procedure types can each pass and still disagree.

Iteration in a model

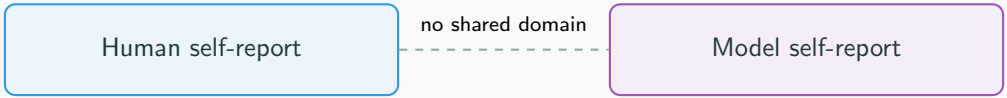
Chang's method, applied to concept injection:



This is the thermometer story again: the starting standard is accepted without proof, and what justifies it later is the record of corrections it made possible.

Extending “introspection” to models

Carrying “introspection” from people to models is a metrological extension.



- **Existence** (Michel): does the model have the state the report is about?
- **Overlap**: human and model self-report share no domain, so neither overlap condition can do any work.
- **Conformity** (Chang) carries the weight: the new standard should agree with whatever meaning “introspection” already has when applied to models.

Where the analogy stops

- **No guaranteed convergence.** Chang's iterations ran for two centuries across many laboratories and converged on a quantity that thermodynamics later explained. Nothing guarantees that introspection measures converge on one quantity. Tal's individuation problem allows that they measure several quantities or none.
- **No trusted starting point.** Sensation of hot and cold had a range in which it was trusted. No such starting principle exists yet for model self-report.
- **No background theory.** Tal's cases, such as temperature and frequency, have theories that fix how they relate to other quantities. For internal states of a model there is none. Tal allows pre-theoretical assumptions in its place.
- **No independent standards.** Tal's standards are kept by institutions independent of the instruments they calibrate. Here the researchers construct the ground truth, and a judge model may score the replies.

Summary

Idea	Question for an introspection experiment	Card stage
Indication, outcome	What is read off the reply, through which model?	Instrument
Fixed points	Does the injected state vary with layer or strength?	Calibration
Kinds of calibration	Is the report pathway modeled, or only reply rates?	Calibration
H1–H3	Why do report and probe disagree?	Comparability
Comparability	Do runs of one procedure agree about one state?	Comparability
Robustness condition	Do diverse procedures agree within uncertainty?	Comparability
Metrological extension	Does the model have the state the report is about?	Validity conditions

Reading

- Chang, *Inventing Temperature* (2004): nomic measurement, comparability, single value, epistemic iteration, metrological extension.
- Tal, Measurement in Science (SEP, 2020): the two-level account; coordination and nomic measurement named together.
- Tal, Calibration: Modelling the measurement process (2017): three kinds of calibration; the generalized robustness condition.
- Tal, Individuating quantities (2019): H1–H3; unity through idealization.
- Tal, How does measuring generate evidence? (2016): outcomes as evidence through epistemic security.
- Michel, Calibration in Consciousness Science (2021): calibration against an inaccurate procedure.
- Michel, Validity drifts in psychiatric research (2024): existence condition; validity drift.